RaceMED: A Race-Aware Approach to Accurate and Fair Medication Recommendation

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Abstract

Clinical evidence indicates that (1) disease prevalence differs across racial groups and (2) medication prescriptions for identical diagnoses and procedures may vary across racial groups. However, existing medication recommendation methods have *not* integrated these racial properties into the design of their encoder and predictor architectures. To address this gap, we propose **RaceMED**, which incorporates properties (1) and (2) through a *dual-branch encoder composed of race-specific and race-general branches* that capture idiosyncratic features unique to each racial group and general patterns shared across all patients, and a predictor based on *race-aware attention* that restricts cross-patient visit references to patients from the same racial group, preventing inappropriate medication transfer across racial groups. Extensive experiments demonstrate that RaceMED consistently outperforms ten state-of-the-art competitors, achieving up to 12.24% improvement in accuracy while reducing performance disparities across racial groups by up to 31.77%, thereby improving fairness. These findings demonstrate that explicitly modeling racial properties is essential for improving both accuracy and fairness, a dimension that has been largely underexplored in the medication recommendation domain.

CCS Concepts

• **Information systems** → Recommender systems.

Keywords

medication recommendation; racial properties; fairness

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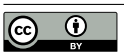

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1 Introduction

Prescribing appropriate medications for patients with complex health conditions is a time-consuming and challenging task, even for experienced doctors [12, 31]. To alleviate this burden, *medication recommender systems* have emerged to support doctors in making clinical decisions. These systems aim to recommend medications appropriate for a patient at her current hospital visit with ensuring safety by reducing Drug-Drug Interactions (DDI) that can cause fatal side effects [11, 12, 20, 31, 33, 34, 36].

Medication recommender systems can be categorized into *instance-based methods* [3, 25, 35, 36] and *longitudinal-based methods* [2, 11, 12, 20, 31, 33, 34]. Instance-based methods recommend medications by utilizing only the diagnoses and procedures given at the patient's current visit. However, they are limited in capturing dynamic health shifts of the patient, as they rely solely on the current visit record [11, 12, 20, 31, 33]. To overcome this limitation, longitudinal-based methods leverage the patient's past visit records in addition to her current visit record. They generally consist of two components: an *encoder* and a *predictor*. In the encoder, they typically employ Recurrent Neural Networks (RNN) [19] or Transformer-based models [24] to encode diagnoses and procedures from both past and current visits into **patient representations** that capture longitudinal changes in the *patient's health status*. In the predictor, utilizing the obtained representations, they predict medications by considering two factors: **(a)** the medications prescribed solely based on the patient's health status at the current visit and **(b)** the medications prescribed at each of her past visits [11, 12, 27, 31]. When considering factor (b), recent longitudinal-based methods evolved their predictors to incorporate medications prescribed at other patients' visits showing health status similar to the target patient's current one, instead of being limited to the target patient's own visits [11, 29, 30]. Consequently, longitudinal-based methods demonstrated superior recommendation accuracy compared to instance-based methods. In this paper, we thus focus on *longitudinal-based methods*.

Limitations. Recent clinical and epidemiological studies have reported **two properties**: **(1)** *differences in disease (in our case, diagnoses and procedures) prevalence exist across racial groups* [9, 14, 16],

and **(2)** *medication prescriptions for identical diagnoses and procedures vary across racial groups* [4, 10]. This suggests that medication recommendation models should consider racial properties (1) and (2) when recommending medications for individual patients. However, existing studies have never considered these racial properties (1) and (2) in designing their encoder and predictor architectures.

Encoder Aspect. Despite property **(1)**, existing models process each patient case from *all* racial groups using a *single* encoder. Since a single encoder minimizes the total loss while sharing parameters across racial groups, it learns patient representations optimized for general diagnostic and procedural patterns that appear frequently in the data, *regardless of* racial nuances. Consequently, this results in *patient representations that are biased towards a specific racial group*, and *entangled* in the latent space *without racial distinction.*

Predictor Aspect. Despite property **(2)**, existing methods utilize factor **(b)**, *i.e.*, medications prescribed at past visits, by retrieving the *all* patients' visits showing health status similar to that of the target patient *without* racial distinction. Thus, the predictor could select *inappropriate patients of other racial groups as reference targets*, who appear similar in representation to her but have been prescribed medications that are *unsuitable* for her due to racial differences.

Taken together, we claim that existing medication recommendation models *overlook racial properties* (1) *and* (2), and their encoder and predictor architectures have limitations in incorporating these properties. As a result, their recommendations could be biased towards specific racial groups, which not only (i) causes *unfair performance disparities across racial groups* but also (ii) *compromises the overall recommendation accuracy* of the models.

Preliminaries. To validate our claim, we conducted preliminary experiments utilizing two state-of-the-art models, RAREMed (SIGIR'24) [36] and HI-DR (AAAI'25) [11], on the widely adopted MIMIC-III [8] dataset; accuracy was evaluated using the metric Jaccard.[1] Please refer to Section 4 for further experimental details. Figure 1 represents the average accuracy across all patients with a dotted line, and the average accuracy disaggregated by racial group with bars. We observe that both models exhibit high accuracy for some racial groups while showing lower accuracy for others, with a accuracy disparity of up to 5.25% across racial groups. When we consider that recent state-of-the-art models outperform their best competitors by an average 5.42% on MIMIC-III in terms of Jaccard in their original papers [2, 11, 12, 20, 31, 33, 34, 36], this result indicates that some racial groups suffer from lower accuracy in medication recommendation leaving the patients in these groups with accuracy levels equivalent to outdated models. For comparison, we present the overall accuracy of our proposed model with a solid line in Figure 1, which provides highly accurate recommendation overall by incorporating racial properties (1) and (2). Remarkably, our model demonstrates accuracy improvement of at least 12.24% over state-of-the-art models. Observed accuracy disparities across racial groups in existing models, together with their lower accuracy, support our claim that overlooking racial properties leads to unfair and less-accurate recommendations. Motivated by these findings,

[1]Similar results were observed on the MIMIC-IV [7] dataset and with other accuracy metrics (PRAUC and F1).

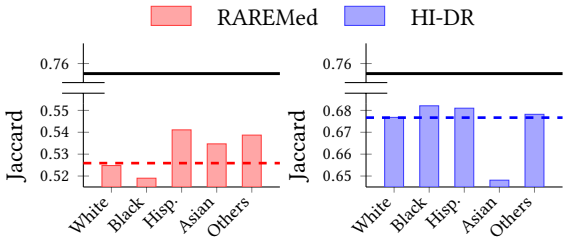


Figure 1: Recommendation accuracy of RAREMed and HI-DR in Jaccard across different racial groups on MIMIC-III. Bars and dotted lines represent the race-specific and overall accuracy, respectively, while the solid line does the overall accuracy of the proposed RaceMED across all racial groups.

we propose **RaceMED**, a novel medication recommendation framework designed to provide accurate and fair recommendations for patients across all racial groups.

To incorporate property **(1)**, the *encoder of* RaceMED utilizes a **dual-branch architecture** to capture multifaceted clinical features. A race-general branch extracts prevalent diagnostic and procedural patterns (**race-general patterns**, for short) *common* across *all* racial groups, while a race-specific branch isolates idiosyncratic patterns (**race-specific patterns**, for short) *unique* to *each* racial group. This process yields distinct race-general and race-specific representations for each patient. Then, RaceMED fuses these representations into a *final patient representation that captures both race-general and race-specific patterns*, resulting in patient representations that are uniformly distributed in the latent space while being distinctly clustered by racial groups (see RQ4 in Section 4).

To reflect property **(2)** in referencing *factor* (b) (*i.e.*, the medications prescribed at past visits), the *predictor of* RaceMED introduces a **race-aware attention mechanism**; specifically, within a latent space where race-specific patterns are encoded, it considers the past visit records of the patients *only* from the *same* racial group as the target patient. This mechanism *prevents* suboptimal recommendations stemming from *spurious latent similarity*, where patients from different racial groups appear proximal in the latent space despite requiring divergent therapeutic regimens due to race-specific physiological factors.

Consequently, RaceMED (i) ensures fairness for all racial groups by reducing accuracy disparities among them, and (ii) achieves higher accuracy by providing appropriate medication recommendations for each racial group (see RQs 1 and 2 in Section 4).

Our contributions are summarized as follows:

- **Important Observations**: We observed that existing medication recommenders overlook racial properties (1) and (2), which not only causes unfair performance disparities across racial groups but also compromises overall accuracy. To the best of our knowledge, this work is the *first attempt* to address the issue of fairness across racial groups in the field of medication recommendation.
- **Novel Framework**: We propose **RaceMED**, a novel medication recommendation framework comprising an encoder with race-general and race-specific branches to incorporate property (1), and a predictor with race-aware attention to reflect property (2).
- **Extensive Experiments**: We demonstrate the effectiveness of RaceMED through extensive experiments on real-world datasets.

RaceMED outperforms the best competitors by up to 12.24% in terms of accuracy and by up to 31.77% in terms of fairness.

2 Related Work

Existing studies on medication recommendation have proposed various techniques, primarily focusing on improving the accuracy and safety of predicted medications. GAMENet [20] uses an RNN to process the patient's visit records from the past to the current visits, capturing longitudinal changes in the patient's health status. Additionally, it constructs Electronic Health Records (EHR) and DDI graphs, modeling medications as nodes and representing their co-prescriptions and DDI relations as edges. By applying Graph Convolutional Networks (GCN) [13] to EHR and DDI graphs, GAMENet incorporates complex correlations between medications into medication representations. As a result, GAMENet achieves medication recommendation that ensures not only accuracy but also safety. SafeDrug [33], MoleRec [34], and SubRec [2] model the chemical structures of medications as graphs and apply GCN to obtain medication representations reflecting their intrinsic chemical properties. Through this approach, they aim to further improve accuracy and safety. VITA [12] and HI-DR [11] selectively utilize only the past visits relevant to the patient's current health status. This avoids the misleading from the irrelevant past visit records, thereby improving the accuracy substantially.

Meanwhile, with the growing interest in fairness in the recommendation domain, studies considering fairness have emerged in the field of medication recommendation. Specifically, RAREMed [36] addresses the data sparsity inherent in the long-tail diagnosis distribution by employing self-supervised pre-training to capture the latent semantic interrelations among various clinical codes. Consequently, it mitigates accuracy degradation for patients with rare diagnoses in terms of fairness.

Discussions. Except for RAREMed, most medication recommendation models are designed primarily to focus on improving accuracy and safety, without considering fairness. To achieve fair and accurate medication recommendation across racial groups, we highlight that race is a clinically critical factor associated with differences in disease prevalence (property (1)) and medication prescription distributions (property (2)). Thus, we claim that explicitly incorporating racial properties (1) and (2) into medication recommendation models is key to achieving both fairness and accuracy. Although RAREMed considers fairness, it does *not* focus on fairness across racial groups, and thus does not explicitly incorporate these properties. As a result, our experimental results show that RAREMed exhibits suboptimal performance in terms of both fairness and accuracy across racial groups (see RQs 1 and 2 in Section 4). Therefore, there is a need for a *novel* learning approach designed for the medication recommendation domain to provide fair and accurate outcomes by addressing racial properties (1) and (2).

3 Proposed Framework: RaceMED

3.1 Preliminaries

Definition 1: Patient Health Records. The health records of a patient $x \in \mathbf{X}$ consist of a sequence of visits, denoted by $\mathcal{V}_x = [\mathcal{V}_x^1, \ldots, \mathcal{V}_x^T]$, where $\mathbf{X}$ denotes set of all patients and $\mathcal{V}_x^T$ denotes T-th visit record of patient x. Let $\mathcal{D}$, $\mathcal{P}$, and $\mathcal{M}$ denote set of all diagnoses, procedures, and medications, respectively. Then each visit record $\mathcal{V}_x^T$ comprises subsets of $\mathcal{D}$, $\mathcal{P}$, and $\mathcal{M}$. To serve as model inputs, these subsets are represented as multi-hot vectors $\mathbf{d}_x^T \in \mathbb{R}^{|\mathcal{D}|}$, $\mathbf{p}_x^T \in \mathbb{R}^{|\mathcal{P}|}$, and $\mathbf{m}_x^T \in \mathbb{R}^{|\mathcal{M}|}$, respectively.

Definition 2: EHR and DDI Graphs. $\mathcal{G}_{EHR} = (\mathcal{M}, \mathcal{E}_{EHR})$ and $\mathcal{G}_{DDI} = (\mathcal{M}, \mathcal{E}_{DDI})$ denote EHR graph and DDI graph, respectively. Here, $\mathcal{E}_{EHR}$ and $\mathcal{E}_{DDI}$ denote sets of edges between medications, where an edge $(i, j) \in \mathcal{E}_{EHR}$ indicates that medications i and j have been prescribed together at least once regardless of the patient or visit, while $(i, j) \in \mathcal{E}_{DDI}$ indicates that medications i and j can cause side effects when taken together[2].

3.2 Problem Definition

Standard Setting. We aim to recommend the set of accurate medications $\hat{\mathcal{M}}_x^T$ for patient x's current visit at T, given x's past visit records $[\mathcal{V}_x^1, \ldots, \mathcal{V}_x^{(T-1)}]$, the diagnoses $\mathbf{d}_x^T$ and procedures $\mathbf{p}_x^T$ at the current visit, and $\mathcal{G}_{EHR}$ and $\mathcal{G}_{DDI}$.

Race-Aware Fairness Setting. A set $\mathbf{X}$ of all patients is partitioned into R disjoint racial groups $\{\mathbf{X}_1, \ldots, \mathbf{X}_R\}$, where R denotes number of racial categories present in EHR. Every patient $x \in \mathbf{X}$ belongs to a single racial group $\mathbf{X}_r$ based on their race r. The goal is to make our RaceMED provide medication recommendations that ensure its accuracy equitable over all racial groups. The racial categories and the fairness metric are described in detail in Section 4.

3.3 Key Components

RaceMED consists of two components, an encoder and a predictor, and its overall flow is presented in Figure 2.

3.3.1 Encoder. In RaceMED, the encoder is composed of a *race-specific branch* and a *race-general branch*. These branches jointly produce a patient representation $\mathbf{q}_x^T \in \mathbb{R}^{dim}$ that captures the *health status* of the patient x at the current visit at T, where dim denotes dimensionality of the representation. Specifically, to address property (1) regarding the disparate prevalence of diagnoses and procedures across racial groups, RaceMED employs the race-specific branch to learn a race-specific patient representation $\mathbf{q}_{x,specific}^T \in \mathbb{R}^{dim}$. This branch captures idiosyncratic patterns, *i.e.*, *race-specific patterns*, that are unique to each racial group. Meanwhile, RaceMED learns a race-general patient representation $\mathbf{q}_{x,general}^T \in \mathbb{R}^{dim}$ via the race-general branch, which captures prevalent diagnostic and procedural patterns, *i.e.*, *race-general patterns*, across all racial groups. Then, RaceMED *fuses* $\mathbf{q}_{x,specific}^T$ and $\mathbf{q}_{x,general}^T$ to obtain the final patient representation $\mathbf{q}_x^T$. We next explain this process in detail using a patient x belonging to a racial group $\mathbf{X}_r$ (*i.e.*, $x \in \mathbf{X}_r$) as an example.

Race-Specific Branch. Recent studies [11, 12] demonstrate that learning a patient representation by selectively leveraging only the past visits relevant to the patient's current visit (rather than using all past visits) and accounting for their degree of relevance better captures the patient's current health status. RaceMED follows the same design principle. Specifically, RaceMED first obtains a representation $\mathbf{v}_x^t \in \mathbb{R}^{dim}$ for every visits $t \in \{1, \ldots, T\}$ of patient $x \in \mathbf{X}_r$ by concatenating ($[\|]$) the diagnoses d_x^t and procedures p_x^t given at each visit, and feeding them into a representation layer:

[2]Following [11, 12, 31], we define medication pairs belonging to the top-40 severity DDI types extracted from [21] as having a relationship that can cause side effects.

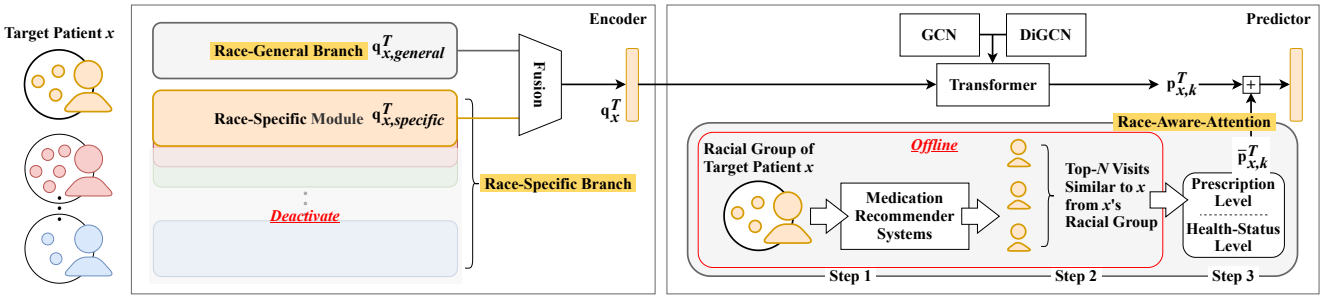


Figure 2: Overview of RaceMED, which consists of two components: an encoder with a dual-branch architecture comprising race-general and race-specific branches, and a predictor based on race-aware attention.

$\forall t \in \{1, \ldots, T\}$,

$$\mathbf{v}_x^t = [\mathbf{d}_x^t \| \mathbf{p}_x^t]\mathbf{W}_r^e, \tag{1}$$

where $\mathbf{W}_r^e \in \mathbb{R}^{(|\mathcal{D}|+|\mathcal{P}|)\times dim}$ denotes weight matrix of the representation layer for racial group $\mathbf{X}_r$. Then, RaceMED computes the probability $s_x^t \in \mathbb{R}$ that each past visit at t is selected as a visit relevant to the current visit at T by concatenating x's current $\mathbf{v}_x^T$ and each past visit representation $\mathbf{v}_x^t$ and feeding them into a Multi-Layer Perceptron (MLP) layer: $\forall t \in \{1, \ldots, (T-1)\}$,

$$s_x^t = sigmoid([\mathbf{v}_x^T \| \mathbf{v}_x^t]\mathbf{W}_r^s + b_r^s), \tag{2}$$

where $\mathbf{W}_r^s \in \mathbb{R}^{2dim\times 1}$ and $b_r^s \in \mathbb{R}$ denote weight matrix and bias of the MLP layer for racial group $\mathbf{X}_r$, respectively. RaceMED then determines whether to select each past visit at t by concatenating the two probabilities s_x^t and $(1 - s_x^t)$ into a two-dimensional vector, and applying Gumbel-softmax [6, 17] to this vector.

Given the past visits relevant to x's current visit, RaceMED calculates the degree a_x^t of relevance between x's current visit at T and every relevant past visit at t by utilizing an attention network [11, 12]: $\forall \mathbf{v}_x^t \in \{all_relevant_past_visits\} \cup \{\mathbf{v}_x^T\}$,

$$a_x^t = softmax(\frac{\{\mathbf{v}_x^T \mathbf{W}_r^a \mathbf{v}_x^{t\intercal}\}}{\sqrt{dim}}), \tag{3}$$

where $\mathbf{W}_r^a \in \mathbb{R}^{dim\times dim}$ denotes weight matrix of the attention network for racial group $\mathbf{X}_r$. Finally, RaceMED obtains a patient representation $\mathbf{q}_{x,specific}^T$ at the current visit, reflecting patterns specific to x's racial group $\mathbf{X}_r$, by taking a weighted sum of $\mathbf{v}_x^T$ for the current visit and $\mathbf{v}_x^t$ for all past visits that are relevant to the current visit, using the degree of relevance a_x^t:

$$\mathbf{q}_{x,specific}^T = \sum_{\mathbf{v}_x^t \in \{all_relevant_past_visits\} \cup \{\mathbf{v}_x^T\}} a_x^t \mathbf{v}_x^t. \tag{4}$$

Note that the weight matrices and biases correspond to race groups and are used to obtain $\mathbf{q}_{x,specific}^T$. Each of them is trained *exclusively* only with the patients within the *same racial group* $\mathbf{X}_r$ *as target patient* x. This design choice prevents the model from learning the diagnosis and procedure patterns specific to the other racial groups, enabling the model to exclusively learn race-specific patterns for racial group $\mathbf{X}_r$, in accordance with property (1).

In summary, the race-specific branch comprises R race-specific modules, each corresponding to a distinct racial group $\mathbf{X}_r$ and parameterized independently. Importantly, such a multi-module architecture does *not* increase the time complexity of RaceMED in proportion to the number of racial categories R. This is because, in the training and inference processes, RaceMED selectively activates *only* a single module corresponding to the target patient's racial group among the R race-specific modules. Hence, the time complexity of the race-specific branch remains $O(L \cdot dim)$, which is identical to that of the original encoders as used in prior work [11, 12], where L denotes number of patient visits. Please refer to Appendix *A* for details on the time complexity.

Race-General Branch. Since race-general patterns could exist alongside race-specific patterns, RaceMED obtains a race-general patient representation $\mathbf{q}_{x,general}^T$ through the race-general branch. In the race-general branch, RaceMED performs the same procedure as in the race-specific branch (*i.e.*, Eqs. 1–4); however, the weight matrices and biases used in this branch are trained on the data from *all* patients. That is, the parameters of the race-general module are shared by *all* patients, independent of a specific racial group.

Fusion. Finally, RaceMED obtains the final representation $\mathbf{q}_x^T \in \mathbb{R}^{dim}$ of patient x capturing *both* race-specific and race-general patterns by concatenating $\mathbf{q}_{x,specific}^T$ and $\mathbf{q}_{x,general}^T$ and feeding them into an MLP layer[3]:

$$\mathbf{q}_x^T = ReLU([\mathbf{q}_{x,specific}^T \| \mathbf{q}_{x,general}^T]\mathbf{W}^F + b^F), \tag{5}$$

where $\mathbf{W}^F \in \mathbb{R}^{2dim\times dim}$ and $b^F \in \mathbb{R}^{dim}$ denote weight matrix and bias of the MLP layer, respectively.

Significance. Using a *single encoder*, existing medication recommendation studies learn patient representations for *all* patients. To this end, they have typically (i) utilized information from the patient's current and all past visits, and recently, (ii) shifted toward utilizing only the relevant past visits [11, 12]. Moving beyond this visit-level selection approach (ii), RaceMED introduces a *patient-level selection* mechanism that identifies a subset of patients who belong to the same racial group as the target patient to learn her race-specific representations. By combining this race-specific representation with a race-general representation, RaceMED departs from the conventional single-encoder design and establishes a novel dual-branch architecture for patient representation learning. It is worth mentioning that simply injecting race-label information into existing single-encoder methods is *not* sufficient to consistently improve both accuracy and fairness,[4] highlighting the importance of of

[3]We experimented with sum, average, and weighted-sum fusion strategies, but all resulted in lower performance than the current design.

[4]We experimented with HI-DR [11], a state-of-the-art method based on a single-encoder architecture, by injecting a 5D race-label vector corresponding to the five racial groups. As a result, on MIMIC-III [8], accuracy measured by Jaccard increased from 0.6767 to 0.6813, but fairness measured by Δ_{sp} Jaccard worsened from 0.0340 to 0.0536. On MIMIC-IV [7], Jaccard decreased from 0.6414 to 0.6320, while Δ_{sp} Jaccard improved from 0.0203 to 0.0170. RaceMED consistently outperformed this variant in terms of both accuracy and fairness (see Table 1).

RaceMED's dual-branch architecture. This design choice mitigates the indiscriminate entanglement of race-specific and race-general patterns inherent in single-encoder approaches, yielding representations that are more uniformly distributed while exhibiting clear race-specific clustering in the latent space, which will be shown in our experiments (see RQ4 in Section 4). Ultimately, this architecture, which enables the learned representations to incorporate racial properties (1) and (2), leads to improved performance; notably, even the variant of RaceMED that utilizes only the race-specific patient representation outperforms *all* state-of-the-art competitors in terms of accuracy, and incorporating the race-general representation further amplifies this advantage. Detailed discussions are provided for RQs 1 and 3 in Section 4.

3.3.2 Predictor. In the predictor, RaceMED predicts the set of medications $\hat{\mathcal{M}}_x^T$ to be prescribed for patient x belonging to the racial group $\mathbf{X}_r$ (*i.e.*, $x \in \mathbf{X}_r$) at the current visit at T. Here, RaceMED predicts medications one by one until the complete prescription set is generated [11, 12, 31]. For simplicity, we describe the predictor of RaceMED via the process of predicting the k-th ($k \geq 1$) medication. RaceMED first obtains all medication representations $\mathbf{e}_i \in \mathbb{R}^{dim}$ by incorporating relationships between co-prescribed medications and those between medications that cause side effects when taken together. Then, using $\mathbf{e}_i$ and the final patient representation $\mathbf{q}_x^T$ from the encoder, RaceMED obtains (a) $\mathbf{p}_{x,k}^T$ and (b) $\bar{\mathbf{p}}_{x,k}^T$:

- (a) $\mathbf{p}_{x,k}^T \in \mathbb{R}^{dim}$ denotes feature vector for the k-th medication to be prescribed when considering only the health status $\mathbf{q}_x^T$ at patient x's current visit at T.
- (b) $\bar{\mathbf{p}}_{x,k}^T \in \mathbb{R}^{|\mathcal{M}|}$ denotes vector indicating the degree to which each medication is likely to be prescribed as the k-th medication for x, based on medications prescribed at visits from the same racial group $\mathbf{X}_r$ that show a health status similar to $\mathbf{q}_x^T$; this choice accounts for property (2), *i.e.*, different medication prescriptions for identical diagnoses and procedures across racial groups.

Finally, RaceMED predicts the k-th medication by combining (a) $\mathbf{p}_{x,k}^T$ and (b) $\bar{\mathbf{p}}_{x,k}^T$.

Medication Representation. Most existing studies obtain all medication representations $\mathbf{e}_i$ capturing co-prescription and side effect relationships among medications by (i) independently applying two-layer GCNs to $\mathcal{G}_{EHR}$ and $\mathcal{G}_{DDI}$ respectively, with randomly initialized e_i, and (ii) subtracting the representations derived from $\mathcal{G}_{DDI}$ and those derived from $\mathcal{G}_{EHR}$ for each medication i [12, 31]. RaceMED follows the same procedure. However, to more accurately reflect co-prescription relationships between medications, it replaces the originally, unweighted indirected $\mathcal{G}_{EHR}$ with a weighted and directed version of EHR graph that reflects the degree of co-prescription at the medication level and captures the asymmetric relationships between medications; it then applies a two-layer DiGCN [22] on this weighted directed graph, instead of a two-layer GCN, to obtain $\mathbf{e}_i$ [11].[5]

(a) Current Health-Aware Feature Vector for k-th Medication. To predict the k-th ($k > 1$)[6] medication, RaceMED computes (a) $\mathbf{p}_{x,k}^T \in \mathbb{R}^{dim}$ by jointly considering the patient's current health-status representation $\mathbf{q}_x^T$ and the sequence of medications predicted in steps 1 through $(k-1)$:

$$\mathbf{p}_{x,k}^T = softmax\left(\frac{transformer(\mathbf{e}^1, \cdots, \mathbf{e}^{(k-1)}) \odot \mathbf{q}_x^T}{\sqrt{dim}}\right)\mathbf{q}_x^T, \quad (6)$$

where $transformer(\cdot)$ denotes same transformer-based model as used in [11, 12, 31]; $\mathbf{e}^{(k-1)}$ denotes representation of the $(k-1)$-th medication predicted at the patient's current visit; and $\odot$ denotes dot-product.

(b) Current Health-Relevant Past Prescription Score Vector for k-th Medication.

Race-Aware Attention. Subsequently, RaceMED obtains (b) $\bar{\mathbf{p}}_{x,k}^T \in \mathbb{R}^{|\mathcal{M}|}$ via our race-aware attention, as follows:

- **Step 1**. Pre-train an existing medication recommendation model on the data from all patients belonging to patient x's racial group $\mathbf{X}_r$ and obtain pre-trained patient representations.
- **Step 2**. From all visits of all patients in $\mathbf{X}_r$, obtain a set $\mathcal{V}_x^{sim}$ of the top-N visits that exhibit pre-trained patient representation most similar to that of x at the current visit.
- **Step 3**. Obtain $\bar{\mathbf{p}}_{x,k}^T$ from all N visit records in $\mathcal{V}_x^{sim}$ by jointly considering (i) their relevance to x's current health status $\mathbf{q}_x^T$ at the health-status level and (ii) their relevance to $\mathbf{p}_{x,k}^T$ at the prescription level.

Step 1. Specifically, to select visits showing a health status similar to x's current health status $\mathbf{q}_x^T$ among the visits of all patients *belonging to* $\mathbf{X}_r$, RaceMED first needs to obtain patient representations for all such visits. To this end, following [11], RaceMED pre-trains an existing medication recommendation model, rather than using its own encoder to obtain these representations during the training process. Importantly, to account for property (1), *i.e.*, different diagnosis and procedure prevalence across racial groups, RaceMED pre-trains the existing model[7] *exclusively* on the data from the patients *belonging to* $\mathbf{X}_r$ to obtain pre-trained patient representations that capture race-specific patterns, whereas [11] pre-trains the model using the data from *all* patients.

Step 2. Using the pre-trained patient representations, RaceMED constructs the set $\mathcal{V}_x^{sim} = \{\mathcal{V}_*^1, \ldots, \mathcal{V}_*^N\}$ of visits selected from all visits of patients in $\mathbf{X}_r$. Here, $\mathcal{V}_*^1, \ldots, \mathcal{V}_*^N$ denote top-N visits whose pre-trained patient representations exhibit the highest cosine similarity to that of x at the current visit; the subscript $*$ indicates that each visit can belong either to x herself or to another patient within the *same racial group* $\mathbf{X}_r$.

It is worth noting that Steps 1 and 2 are performed *offline* before the training phase of RaceMED; that is, they are *not* executed again during RaceMED's training and inference, and therefore introduce *no* additional computation overhead in either phase.[8] Furthermore, this offline stage can be practical, ensuring reusability for new patients with limited performance degradation.[9]

[5] We further experimented with race-specific weighted directed EHR graphs, as such a design choice may follow property (2); however, this variant led to decreases in both accuracy and fairness, and is therefore omitted from the final model.

[6] For $k = 1$, a randomly initialized <START> token is used [11, 12, 31].

[7] To obtain patient representations that more accurately reflect the patient's current health status, we use VITA [12], which has an enhanced encoder, as the pre-training model. We also examined other models, but accuracy degradation was observed.

[8] On MIMIC-III [8], even including this offline stage, RaceMED's total time is ~245s, comparable to SubRec [2] (~222s). A similar trend was observed on MIMIC-IV [7].

[9] On MIMIC-III [8], we held out 1–20% of patients from the training set during pre-training and evaluated RaceMED under this setting afterward. Even at the 20% setting, RaceMED maintained a Jaccard of 0.7060 and Δ_{sp} Jaccard of 0.0211, showing *no*

Step 3. RaceMED obtains $\bar{\mathbf{p}}_{x,k}^T$ by considering the degree of relevance between x's current visit at T and each visit $\mathcal{V}_*^n \in \mathcal{V}_x^{sim}$ (where $n \in \{1, \ldots, N\}$) at both the health-status level $\sigma^n \in \mathbb{R}$ and the prescription level $\mu_i^n \in \mathbb{R}$ [11, 12, 31]. First, σ^n denotes how relevant the health status at x's current visit is to that at $\mathcal{V}_*^n$. Here, although the pre-trained patient representations could be directly used to represent health status, they capture only race-specific patterns (see Step 1). Therefore, to additionally capture race-general patterns, RaceMED obtains the health status $\mathbf{q}_*^n$ at $\mathcal{V}_*^n$ using its encoder and employs it to compute σ^n. Formally, RaceMED computes σ^n using an attention network [11, 12]: $\forall \mathbf{q}_*^n, \forall n = \{1, \ldots, N\}$,

$$\sigma^n = \frac{exp(\{\mathbf{q}_x^T \mathbf{W}^\sigma \mathbf{q}_*^{n\intercal}\}/\sqrt{dim})}{\sum_{j=1}^{N} exp(\{\mathbf{q}_x^T \mathbf{W}^\sigma \mathbf{q}_*^{j\intercal}\}/\sqrt{dim})}, \tag{7}$$

where $\mathbf{W}^\sigma \in \mathbb{R}^{dim \times dim}$ denotes weight matrix of the attention network. Second, μ_i^n denotes how relevant medication i prescribed at $\mathcal{V}_*^n$ is to the feature vector $\mathbf{p}_{x,k}^T$ for the k-th medication to be prescribed at x's current visit at T [11, 12, 31]: $\forall i \in \{medications$ ($meds.$, for short)$_prescribed_at\ \mathcal{V}_*^n\}, \forall n \in \{1, \ldots, N\}$,

$$\mu_i^n = \frac{exp(\{\mathbf{e}_i^n \odot \mathbf{p}_{x,k}^T\}/\sqrt{dim})}{\sum_{j \in \{meds._prescribed_at\ \mathcal{V}_*^n\}} exp(\{\mathbf{e}_j^n \odot \mathbf{p}_{x,k}^T\}/\sqrt{dim})}, \tag{8}$$

where e_i^n denotes representation of medication i prescribed at $\mathcal{V}_*^n$. By combining σ^n and μ_i^n, RaceMED obtains $\bar{\mathbf{p}}_{x,k}^T \in \mathbb{R}^{|\mathcal{M}|}$:

$$\bar{\mathbf{p}}_{x,k}^T = \sum_{n=1}^{N} \sigma^n \mu^n \ \ (\mu^n \in \mathbb{R}^{|\mathcal{M}|}),$$
$$\text{where } \forall i \in \mathcal{M},\ \mu^n = \begin{cases} \mu_i^n, & \text{if } i \in \{meds._prescribed_at\ \mathcal{V}_*^n\}, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Prediction. Finally, RaceMED computes the prediction score $\hat{\mathbf{p}}_{x,k}^T \in \mathbb{R}^{|\mathcal{M}|}$ for each medication to be prescribed as the k-th medication at patient x's current visit by integrating the two vectors (a) $\mathbf{p}_{x,k}^T$ and (b) $\bar{\mathbf{p}}_{x,k}^T$, and predicts the k-th medication [11, 12, 31]:[10]

$$argmax_{k \in \{1,\ldots,|\mathcal{M}|\}}(\hat{\mathbf{p}}_{x,k}^T),$$
$$\text{where } \hat{\mathbf{p}}_{x,k}^T = \lambda_k \{softmax(\mathbf{p}_{x,k}^T \mathbf{W}^p + b^p)\} + (1 - \lambda_k)\bar{\mathbf{p}}_{x,k}^T; \tag{10}$$

λ_k denotes learnable parameter that controls the balance between $\mathbf{p}_{x,k}^T$ and $\bar{\mathbf{p}}_{x,k}^T$; $\mathbf{W}^p \in \mathbb{R}^{dim \times |\mathcal{M}|}$ and $b^p \in \mathbb{R}^{|\mathcal{M}|}$ denote weight matrix and bias, respectively.

Significance. Predictors in existing medication recommender systems that reference medications prescribed at past visits have evolved through three approaches: (i) referencing all past visits of the target patient; (ii) selectively referencing only her past visits that exhibit health status similar to the current visit; and more recently, (iii) referencing visits from all patients (including the target patient) that exhibit health status similar to that of her current visit [11, 29, 30]. While approach (iii) achieves substantial gains in accuracy by leveraging cross-patient records, they expand the reference scope *without* accounting for property (2). As a result, the reference pool may include those visits from patients belonging to *other racial groups* whose representations appear similar in the latent space but whose prescribed medications may be *inappropriate* for the target patient due to *racial differences*. Such visits can be regarded as *false neighbors*. To address this issue regarding the reference scope, RaceMED introduces a novel race-aware attention mechanism. This mechanism preserves the performance benefits of cross-patient referencing while *restricting* the reference pool to the patients belonging to the *same racial group* as the target patient (*i.e.*, Step 2). By architecturally excluding false neighbors, RaceMED ensures that referenced visits are *not only* similar in representation *but also* consistent with prescription patterns suitable for the target patient, thereby improving the appropriateness of recommended medications. Furthermore, race-aware attention reflects property (1) at the pre-training stage (*i.e.*, Step 1) by learning patient representations exclusively from the patients *within* the same racial group. Consequently, the learned representations capture race-specific disease prevalence characteristics, allowing visit retrieval to compare patients based on diagnosis and procedure distributions that are more representative for that group. Taken together, the constrained reference scope and race-aware pre-training jointly address the limitations of unconstrained cross-patient attention and lead to substantial improvements in both recommendation accuracy and fairness. Detailed analyses are provided for RQ3 in Section 4.

3.3.3 Training. We learn all medication representations e_i and the other learnable parameters of RaceMED by minimizing the cross-entropy loss on the predicted medications [11, 12, 31]:

$$\mathcal{L} = -\sum_{x \in \mathbf{X}} \sum_{t=1}^{T} \sum_{i=1}^{|\mathcal{M}|} m_{x,i}^t log(\hat{\mathbf{p}}_{x,k,i}^t), \tag{11}$$

where $m_{x,i}^t \in \mathbb{R}$ denotes whether medication i was actually prescribed at visit t of patient x, taking the value 1 if prescribed and 0 otherwise. Note here that DDI loss regularization [20] can be easily incorporated when stricter DDI rate control is required. However, we use the setting without DDI loss as the default to preserve realistic prescription patterns, as stricter DDI rate control can reduce accuracy. This trade-off is analyzed in RQ1 in Section 4.

4 Evaluation

In this section, we evaluate the effectiveness of RaceMED by addressing the following five key research questions (RQs):

- **RQ1**: Does RaceMED outperform state-of-the-art medication recommendation models in terms of both *accuracy* and *safety*?
- **RQ2**: Does RaceMED improve *fairness* by reducing performance disparities across racial groups compared with competitors?
- **RQ3**: Do RaceMED's (i) dual-branch encoder and (ii) race-aware attention mechanism contribute to its improvements in accuracy, safety, and fairness?
- **RQ4**: Do the patient representations learned by RaceMED form distinct clusters across racial groups in the latent space?
- **RQ5**: Is RaceMED robust across race-related scenarios?

4.1 Experimental Settings

4.1.1 Datasets. We conducted experiments on MIMIC-III [8] and MIMIC-IV [7], which are real-world EHR datasets widely used in the medication recommendation domain; for each dataset, we filtered out those patients with only a single visit, as well as the diagnoses, procedures, and medications associated only with them [2, 11, 12, 20, 26, 27, 31–34, 36]. Please refer to Appendix *B* for detailed statistics of the two datasets. As mentioned earlier, we aim to provide accurate

substantial collapse from the original performance, while *still outperforming* the state-of-the-art HI-DR [11] in accuracy with comparable fairness (see Table 1).

[10]The prediction process continues until the <END> token is generated [11, 12, 31].

Table 1: Accuracy, safety, and fairness of ten competitors and RaceMED. The improvements of RaceMED over the best competitors are all statistically significant with p-value ≤ 0.01.

Dataset	MIMIC-III (Ground-Truth DDI Rate: 0.0815)							
Metrics	**Jaccard (↑)**	**PRAUC (↑)**	**F1 (↑)**	**DDI rate (↓)**	**Δ_{sp} Jaccard (↓)**	**Δ_{sp} PRAUC (↓)**	**Δ_{sp} F1 (↓)**	**Δ_{sp} DDI rate (↓)**
RETAIN (NeurIPS'16)	0.4193 ± 0.0025	0.6652 ± 0.0015	0.5804 ± 0.0023	0.0867 ± 0.0007	0.0435 ± 0.0072	0.0637 ± 0.0085	0.0436 ± 0.0087	0.0067 ± 0.0029
LEAP (KDD'17)	0.3933 ± 0.0016	0.5724 ± 0.0032	0.5507 ± 0.0018	**0.0581 ± 0.0025**	0.0239 ± 0.0142	0.0424 ± 0.0119	0.0264 ± 0.0141	0.0126 ± 0.0020
GAMENet (AAAI'19)	0.4411 ± 0.0049	0.6861 ± 0.0032	0.5986 ± 0.0049	0.0740 ± 0.0008	0.0380 ± 0.0009	0.0647 ± 0.0072	0.0372 ± 0.0030	0.0119 ± 0.0058
SafeDrug (IJCAI'21)	0.5074 ± 0.0030	0.7592 ± 0.0025	0.6641 ± 0.0027	0.0611 ± 0.0005	0.0438 ± 0.0126	0.0491 ± 0.0126	0.0384 ± 0.0117	**0.0036 ± 0.0022**
COGNet (WWW'22)	0.5296 ± 0.0018	0.7679 ± 0.0011	0.6828 ± 0.0016	0.0819 ± 0.0009	0.0393 ± 0.0062	0.0282 ± 0.0032	0.0324 ± 0.0049	0.0083 ± 0.0085
MoleRec (WWW'23)	0.5324 ± 0.0017	0.7748 ± 0.0006	0.6860 ± 0.0016	0.0729 ± 0.0009	0.0617 ± 0.0202	0.0743 ± 0.0116	0.0558 ± 0.0161	0.0061 ± 0.0019
RAREMed (SIGIR'24)	0.5259 ± 0.0008	0.7915 ± 0.0012	0.6801 ± 0.0007	0.0715 ± 0.0008	0.0211 ± 0.0025	0.0223 ± 0.0063	0.0192 ± 0.0022	0.0141 ± 0.0022
VITA (AAAI'24)	0.5242 ± 0.0010	0.7638 ± 0.0018	0.6767 ± 0.0010	0.0809 ±0.0020	0.0234 ± 0.0049	0.0303 ± 0.0021	0.0232 ± 0.0025	0.0043 ± 0.0012
SubRec (NeurIPS'25)	0.5320 ± 0.0006	0.7735 ± 0.0015	0.6855 ± 0.0006	0.0743 ± 0.0003	0.0412 ± 0.0046	0.0463 ± 0.0113	0.0353 ± 0.0049	0.0045 ± 0.0013
HI-DR (AAAI'25)	0.6767 ± 0.0011	0.8399 ± 0.0006	0.7797 ± 0.0010	0.0815 ± 0.0001	0.0340 ± 0.0012	0.0250 ± 0.0012	0.0242 ± 0.0011	0.0050 ± 0.0001
RaceMED	**0.7595 ± 0.0009**	**0.8803 ± 0.0007**	**0.8334 ± 0.0009**	0.0825 ± 0.0009	**0.0188 ± 0.0027**	**0.0171 ± 0.0036**	**0.0131 ± 0.0022**	0.0074 ± 0.0008
Improvements (%)	**12.24**	**4.81**	**6.89**	-	**10.90**	**23.32**	**31.77**	-

and fair recommendations across racial groups by considering racial properties (1) and (2). To this end, we utilized the race information of patients included in MIMIC-III and MIMIC-IV;[11] their racial distributions are shown in Figure 3. The distribution of patients across racial groups is *imbalanced*, which supports the necessity of analyzing fairness across racial groups in this study.

4.1.2 Competitors. We compare RaceMED with ten state-of-the-art medication recommendation models [1, 2, 11, 12, 20, 31, 33–36].

4.1.3 Metrics. To comprehensively evaluate accuracy, safety, and fairness, we employ the following metrics:

- **Accuracy**: Jaccard, PRAUC, and F1, as in prior work [2, 11, 12, 20, 26, 27, 31–36].
- **Safety**: DDI rate [20], which measures the proportion of medication pairs that can cause side effects among all pairs in the predicted medications. A *lower* DDI rate indicates that the model predicts *safer* medication combinations.
- **Fairness**: To quantify performance disparities across racial groups, we adapt Statistical Parity (Δ_{sp}) to our setting, a metric widely used in fairness research [15, 18, 28]:

$$\Delta_{sp} = \max_{r \in R} Performance(\mathbf{X}_r) - \min_{r \in R} Performance(\mathbf{X}_r). \quad (12)$$

That is, Δ_{sp} measures the difference between the maximum and minimum values of a performance metric observed across all racial groups $\mathbf{X}_r$. A *lower* Δ_{sp} value indicates a *fairer* model that provides *more-equitable* performance across racial groups.

4.1.4 Implementation Details. We randomly split all patients in each dataset into training (4/6), validation (1/6), and test (1/6) sets, following [2, 11, 12, 20, 31, 33, 34, 36].

For a fair comparison, we set the representation dimension (*i.e.*, *dim*) to 64 for all competitors, including RaceMED [1, 2, 11, 12, 20, 31, 33–36]. Furthermore, we carefully tuned the hyper-parameters of all models via grid search. Specifically, in RaceMED, we used five racial groups (*i.e.*, $R = 5$) from the EHR race annotations: White, Black, Hispanic, Asian, and Others, following the racial categories used in BBC presidential election statistics[12]. We set the key hyper-parameters of RaceMED as follows: learning rate=1×10^{-3}, batch size=16, Gumbel-softmax temperature=0.8, and N=3.

[11] Please refer to Appendix *C* for detailed information on racial categories.
[12] https://www.bbc.com/news/election-us-2016-37922587

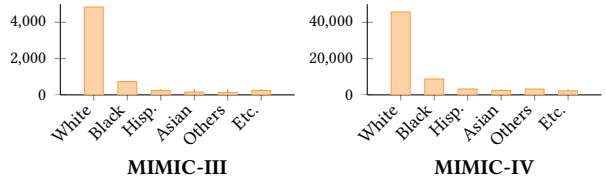


Figure 3: Distribution of patients across racial groups on MIMIC-III and MIMIC-IV. The x-axis denotes racial groups, and the y-axis shows the number of patients.

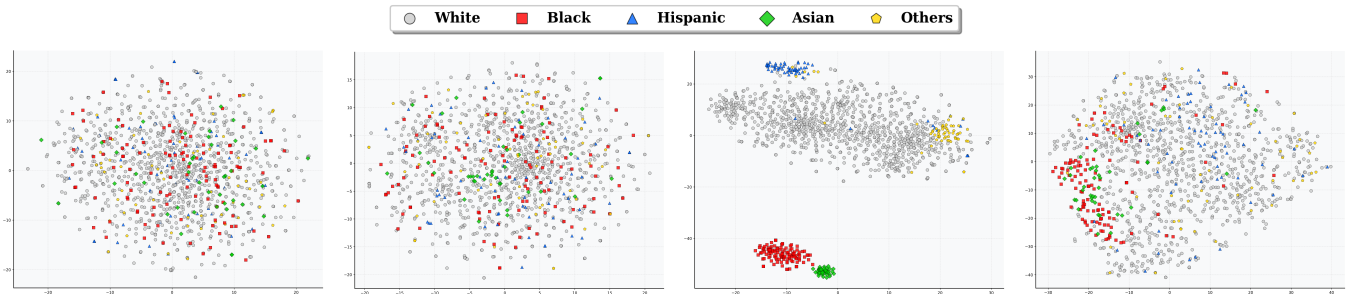
4.2 Results and Analysis

We report the mean values over five independent runs for all experimental results, with standard deviations provided in Table 1; all improvements are *statistically significant* with a p-value ≤ 0.01. For clarity, we highlighted the best performance in each column (*i.e.*, metric) in **bold** and the second-best with an underline. Due to space limitations, the results on MIMIC-IV are provided in Appendix *D* as they exhibit *similar tendencies* to those on MIMIC-III.

RQ1: Accuracy and Safety Comparison with Ten Competitors. Table 1 presents the accuracy and safety results of RaceMED and all competitors on MIMIC-III, revealing the following key findings.

Accuracy. RaceMED achieves the best accuracy, *significantly* outperforming *all* competitors for *all* three accuracy metrics. Specifically, RaceMED outperforms the best competitor HI-DR by up to 12.24% in terms of Jaccard. This improvement is substantial, given that existing methods outperform their best competitors by an average of 5.42% on MIMIC-III in terms of Jaccard in their original papers [1, 2, 11, 12, 20, 31, 33–36]. This suggests that our idea of explicitly incorporating racial properties into the recommendation process plays a *pivotal role* in enhancing overall accuracy.

Safety. In terms of DDI rate, RaceMED does not achieve the best performance (*i.e.*, the safest result). However, it is worth noting that, in the ideal case where a model's accuracy approaches 100% and perfectly mimics a human doctor's prescription, the model's DDI rate will inevitably converge to the ground-truth DDI rate. By achieving the highest accuracy while maintaining DDI rate (0.0825) close to the ground-truth (0.0815), RaceMED demonstrates that it has learned doctors' prescribing patterns more accurately than competitors without significantly compromising safety compared



(a) HI-DR (b) Race-General (c) Race-Specific (d) Final

Figure 4: Visualization of patient representations learned by (a) HI-DR, (b) RaceMED's race-general branch, and (c) RaceMED's race-specific branch, as well as (d) RaceMED's final representations. Colors and markers denote different racial groups.

to actual prescriptions. Nevertheless, recall that DDI loss regularization can be easily incorporated into RaceMED's loss function when stricter DDI rate control is required. Although this setting may deviate from realistic prescription patterns, we evaluated a variant of RaceMED whose loss function incorporates DDI loss regularization to recommend safer medication combinations. When constrained to a lower DDI rate of 0.0749, this variant achieves a Jaccard of 0.6690. This constrained variant remains *comparable* to the best competitor HI-DR in terms of accuracy (0.6690 vs. 0.6767), while being *substantially safer* (0.0749 vs. 0.0815). Moreover, under this lower DDI rate setting, RaceMED achieves DDI rate *comparable* to baselines designed to reduce it, while achieving *much higher* accuracy than those baselines, including GAMENet, MoleRec, RAREMed, and SubRec. These results indicate that RaceMED can explicitly control the DDI rate when stricter safety is required, but we use the setting without DDI loss as the default to preserve realistic prescription patterns *without* artificially distorting them.

RQ2: Fairness Comparison with Ten Competitors. Table 1 presents the performance disparities across racial groups in terms of Δ_{sp} for RaceMED and all competitors. RaceMED achieves the *lowest* Δ_{sp} values across *all* evaluation metrics, *except for* DDI rate, demonstrating its superior fairness over all competitors. Specifically, RaceMED achieves improvements of 10.90% in terms of Δ_{sp} Jaccard compared to the best competitor. This demonstrates that RaceMED ensures consistent recommendation performance across all racial groups *without being biased toward any specific one*, thereby achieving fairness successfully. Importantly, although Δ_{sp} is widely used in fairness research and we already evaluated it over four medication recommendation metrics, we additionally evaluated Equalized Odds (EqOdds) [5] and Equality of Opportunity (EO) [5], which assess group-level fairness in prediction outcomes from different perspectives [5, 15, 18, 28]. Under both EqOdds and EO, RaceMED remains the *fairest* model, showing up to a 447.36% relative improvement in fairness over VITA, the strongest competitor under these metrics. Overall, these findings suggest that explicitly incorporating racial properties into the recommendation process effectively mitigates performance disparities across all racial groups.

RQ3. Ablation Study. To verify the effectiveness of RaceMED's two core ideas (i) the dual-branch encoder composed of race-specific and race-general branches, and (ii) the race-aware attention mechanism, we compare RaceMED with its four variants. For a fair comparison, we keep the same optimal hyper-parameters as RaceMED fixed across all variants.

Table 2: The effects of RaceMED's two core ideas: dual-branch encoder and race-aware attention mechanism

	Gen.	Spe.	S1	S2	Jaccard	DDI rate	Δ_{sp} Jac.
RaceMED	✓	✓	✓	✓	**0.7595**	**0.0825**	**0.0188**
V1	✓	-	✓	✓	0.7375	0.0843	0.0243
V2	-	✓	✓	✓	0.7351	<u>0.0829</u>	<u>0.0234</u>
V3	✓	✓	-	-	0.7237	0.0839	0.0446
V4	✓	✓	-	✓	<u>0.7385</u>	0.0846	0.0275

Encoder Variants V1 and V2. To examine the effect of the dual-branch encoder, we remove each branch individually. **V1** removes the race-specific branch from RaceMED's encoder (*i.e.*, $\mathbf{q}_x^T = \mathbf{q}_{x,general}^T$ in Eq. 5). **V2** removes the race-general branch from RaceMED's encoder (*i.e.*, $\mathbf{q}_x^T = \mathbf{q}_{x,specific}^T$ in Eq. 5).

Predictor Variants V3 and V4. Race-aware attention restricts both the pre-training data in Step 1 and the reference visits in Step 2 to those belonging to the same racial group as the target patient. **V3** removes the restrictions in both Steps 1 and 2 of race-aware attention. In V3, all patients are used for pre-training, and reference visits are selected from patients of all racial groups. This corresponds to the approach adopted in prior work [11]. **V4** removes the restriction in Step 1 of race-aware attention. Patient representations are learned from all patients, while reference visits remain restricted to those from patients belonging to the same racial group as the target patient. We also considered a variant that removes only the restriction on reference visits while preserving pre-training based on race-specific data. However, because race-specific pre-training results in separate latent spaces for each racial group, visits from other racial groups cannot serve as valid reference visits under this design choice. Therefore, this variant is not considered.

Table 2 presents the performance of RaceMED and all its variants in terms of Jaccard, DDI rate, and Δ_{sp} Jaccard.[13] Experimental results show that RaceMED achieves *superior performance* for *all* metrics compared to *all* variants. Furthermore, notably, even *all* variants *consistently* outperform *all* competitors in terms of accuracy (see the Jaccard results in Table 1). These results demonstrate

[13]Due to space limitations, the results for the remaining metrics are reported in Appendix *D*.

that, when individually applied, each of our core ideas is *effective*, when combined, they create a synergistic effect that achieves the *highest* performance. Together, these findings indicate that *architecturally* incorporating racial properties (1) and (2) into both the encoder and predictor is key to mitigating performance disparities across racial groups while improving overall accuracy and safety.

RQ4. Representation Analysis. We visualize the latent space of patient representations using t-SNE [23]. Figure 4 presents the distributions of four types of patient representations: (a) those learned by the best competitor HI-DR, (b) race-general representations learned by RaceMED, (c) its race-specific representations, and (d) its final representations. Patients are shown using different colors and markers according to their racial groups. We observe that HI-DR's patient representations are *indiscriminately* entangled across racial groups, *without clear separation* in the latent space. In contrast, RaceMED's race-specific patient representations form *distinct clusters*, corresponding to different racial groups, indicating that RaceMED effectively captures race-specific patterns and disentangles them across racial groups. RaceMED's race-general patient representations exhibit an entangled distribution similar to that of HI-DR, reflecting its role in capturing race-general patterns shared across racial groups. The final patient representations of RaceMED display a balanced structure in the latent space, preserving *both* shared patterns across racial groups and distinctions specific to each racial group; this balanced structure explains the best performance in terms of *both* accuracy and fairness (see Table 2).

RQ5. Robustness. Since RaceMED uses race-aware constraints in both the encoder and predictor, one may be concerned that such a design could reduce the effective training or reference data available for underrepresented racial groups, thereby limiting representation learning or generalization for those groups. To address this concern, we evaluated whether RaceMED remains robust across race-related scenarios, including *underrepresented racial groups*, *mixed-heritage patients*, and *noisy race labels*.

First, RaceMED does *not* simply degrade for smaller racial groups. On MIMIC-III, we conducted an evaluation on extreme micro-populations with only 1–4 patients, such as American Indian in Table 4. RaceMED achieved stable Jaccard scores of 0.7298–0.7823 for these micro-populations, *far* exceeding the best competitor HI-DR's Jaccard of 0.6767 (see Table 1). This suggests that group size is *not* the primary factor determining RaceMED's performance.

Second, following the race annotations in the EHR datasets, RaceMED uses five mutually exclusive racial categories to capture observed clinical differences, while patients outside these categories, including mixed-heritage patients, are assigned to the *Others* group. This group therefore contains patients from relatively small or heterogeneous racial categories. The performance of the Others group remains *comparable* to the overall average (see Figure 5), suggesting that RaceMED does *not* amplify existing disparities for these minority cases. To further validate this, in an additional evaluation on nine held-out mixed-heritage patients, *i.e.*, the Multi Race Ethnicity group in Table 4, RaceMED achieved a Jaccard of 0.7390 on MIMIC-III, matching the overall average and outperforming HI-DR's Jaccard of 0.6767.

Third, RaceMED is *robust* to noisy race labels. We randomly altered 10–30% of patient race labels to incorrect categories. Even under the extreme 30% noise setting, RaceMED's Jaccard decreased

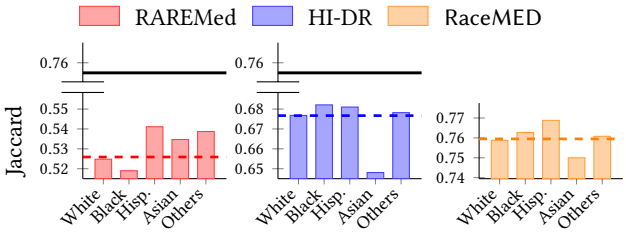


Figure 5: Recommendation accuracy of RAREMed, HI-DR and RaceMED in Jaccard across different racial groups on MIMIC-III. Bars and dotted lines represent the race-specific and overall accuracy, respectively, while the solid line does the overall accuracy of the proposed RaceMED across all racial groups.

only from 0.7595 to 0.7414, while *still* outperforming HI-DR's Jaccard of 0.6767. This indicates that RaceMED does *not* rely on perfectly clean race labels to maintain recommendation performance.

These results align with the design goal of RaceMED's architecture. The encoder fuses race-general and race-specific branches, *preventing* representations from being overly clustered by race or overly dispersed across races. Meanwhile, the predictor's same-race constraint acts as a *safeguard* against inappropriate medication transfer from spurious latent similarity across racial groups. As shown in RQ3, removing the same-race constraint *worsens all metrics*, including Jaccard (0.7595 → 0.7237), DDI rate (0.0825 → 0.0839), and Δ_{sp} Jaccard (0.0188 → 0.0446). These findings suggest that RaceMED's race-aware design does *not* merely reduce usable data for underrepresented groups; rather, it enables controlled race-aware representation learning and referencing.

5 Conclusions

We proposed RaceMED, the *first* medication recommendation framework that explicitly incorporates *racial properties* into both the encoder and predictor architectures. RaceMED models patient representations through a *dual-branch encoder that separately captures race-general and race-specific patterns*, enabling the model to learn both clinical features shared across racial groups and those distinct to each group. On the prediction side, RaceMED *constrains cross-patient medication referencing* to only the patients belonging to the same racial group as the target patient, thereby architecturally preventing inappropriate transfer of prescription patterns across racial groups. Extensive experiments showed that RaceMED *consistently* and *significantly* improved overall accuracy and reduced performance disparities across racial groups without compromising safety. These results indicate that, in medication recommendation, racial properties should not be ignored but explicitly reflected in model design to achieve both high accuracy and fairness.

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Appendix A Time Complexity

In this section, we analyze the time complexity of RaceMED to demonstrate its efficiency. The computational cost of the overall model primarily stems from the encoder, which processes the extensive visit records of patients, and the predictor, which retrieves relevant visits of all patients belonging to $\mathbf{X}_r$. Therefore, we focus our analysis on the time complexity of these two core components.

Encoder. RaceMED's encoder adopts a dual-branch structure, consisting of a race-general branch and a race-specific branch, to learn patient representations. By analyzing the time complexity of each branch, we mathematically demonstrate that the overall model remains linear with respect to the patient visit length. L denotes

patient visit length and dim the representation dimension. The computational cost C for each module is derived as follows:

1. Race-general branch time complexity ($C_{general}$). This branch consists of a single branch designed to learn race-general patterns across all patients. Its complexity is given by:

$$C_{general} = O(L \cdot dim). \tag{13}$$

2. Race-specific branch time complexity ($C_{specific}$). This branch comprises a total of R individual modules, $\{E_1, E_2, \ldots, E_R\}$, to capture race-specific patterns. However, given a patient x with race r, only the single module E_r corresponding to that race is activated during the model inference phase, formulated as:

$$C_{specific} = \sum_{k=1}^{|R|} I(k = r) \cdot O(E_k) = 1 \cdot O(L \cdot dim) = O(L \cdot dim). \tag{14}$$

where I is the indicator function. Consequently, the actual computational cost is equivalent to that of the race-general branch.

3. Fusion layer time complexity (C_{fusion}). The MLP-based fusion layer, which integrates the outputs of the two branches, incurs a constant cost, as it performs matrix operations solely on a fixed dimension dim:

$$C_{fusion} = O(dim^2). \tag{15}$$

Total Complexity (C_{total}). Finally, the total time complexity of RaceMED's encoder is determined by the sum of the costs of the race-general and race-specific branches:

$$C_{total} = C_{general} + C_{specific} + C_{fusion} = 2 \cdot O(L \cdot dim) + O(dim^2). \tag{16}$$

Since the representation dimension dim is considered a relatively small constant compared to the patient visit length L:

$$C_{total} \approx O(L \cdot dim) \tag{17}$$

Consequently, although RaceMED incorporates an additional race-specific branch and a fusion layer compared to existing encoders, the overall computational cost remains linear with respect to the patient visit length L, *i.e.*, $O(L)$. This mathematically demonstrates that the inclusion of additional race-specific branches does not significantly compromise the model's time complexity.

Predictor. As noted in prior work [11], retrieving relevant visits by generating patient representations in real-time and calculating similarity across the entire dataset is a primary factor that exponentially increases model training and inference time. Accordingly, consistent with HI-DR, we adopt a strategy of pre-computing patient representations using a pre-trained medication recommendation model. Consequently, during the actual model training and inference phases, RaceMED incurs no additional computational overhead for representation generation or similarity calculation compared to the backbone model HI-DR. The model simply requires retrieving $\mathbf{V}_x^{sim}$ via a lookup on the pre-stored index with a time complexity of $O(1)$. Therefore, despite performing sophisticated race-aware attention, RaceMED maintains a high level of time efficiency comparable to the backbone model without any additional computational burden.

Appendix B Datasets

To validate the effectiveness of RaceMED, we conduct experiments using real-world EHR datasets, MIMIC-III and MIMIC-IV. The statistics of each dataset are summarized in Table 3.

Table 3: Dataset statistics

Item	MIMIC-III	MIMIC-IV
# of visits / # of patients	15,032 / 6,350	174,512 / 65,484
# of dis. / proc. / med.	1,958 / 1,430 / 131	1,994 / 11,358 / 124
avg. / max # of visits	2.37 / 29	2.66 / 70
avg. / max # of dis. per visit	13.63 / 39	11.58 / 48
avg. / max # of proc. per visit	4.54 / 32	2.75 / 34
avg. / max # of med. per visit	19.20 / 53	12.50 / 65

Table 4: Racial distribution statistics

Race	MIMIC-III		MIMIC-IV	
	Num	%	Num	%
White	4,841	76.24	45,682	69.76
Black	736	11.59	8,736	13.34
Hispanic	241	3.80	3,242	4.95
Asian	157	2.47	2,412	3.68
Others	134	2.11	3,210	4.90
Unknown/Not specified	144	2.27	-	-
Declined to answer	44	0.69	-	-
Unable to obtain	20	0.31	352	0.54
Portuguese	11	0.17	-	-
Multi Race Ethnicity	9	0.14	-	-
Middle Eastern	6	0.09	-	-
American Indian/Alaska Native	4	0.06	150	0.23
American Indian/Alaska Native Federally Recognized Tribe	1	0.02	-	-
Native Hawaiian or Other Pacific Islander	1	0.02	-	-
Caribbean Island	1	0.02	-	-
Unknown	-	-	1700	2.60
Total	**6,350**	**100.0**	**65,484**	**100.0**

Appendix C Race Distribution

We report the racial distribution statistics for the MIMIC-III and MIMIC-IV at Table 4, presenting the patient count and proportion for each race. In this work, patients belonging to minority racial groups, including the original Others group, are collectively classified as Others group.

Appendix D Additional Results for RQs

In this section, we presents the additional experimental results for RQs. The results for RQ1 and RQ2 on MIMIC-III, RQ3 on both MIMIC-III and MIMIC-IV in terms of Jaccard, PRAUC, F1, and their corresponding Δ_{sp}, and RQ4 on MIMIC-IV. Specifically, The results for RQ1 and RQ2 are presented in Table 5, and those for RQ3 are shown in Tables 6 and 7, while the findings for RQ4 are illustrated in Figure 6. All experimental results are statistically significant with p-value ≤ 0.01. As discussed in RQ3 of Section 4, we demonstrate that the encoder composed of race-specific and general branches and the race-aware attention are effective individually, and when combined, they create a synergistic effect to achieve higher performance. This phenomenon is consistently observed across accuracy and fairness metrics, with the exception of DDI rate.

Table 5: Accuracy, safety, and fairness of ten competitors and RaceMED on MIMIC-IV. The improvements of RaceMED over the best competitors are all statistically significant with p-value ≤ 0.01.

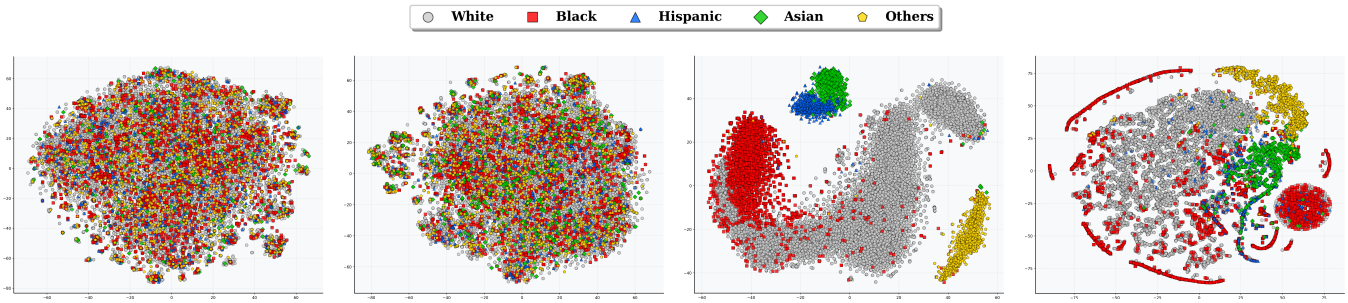
Dataset	MIMIC-IV (Ground-Truth DDI Rate: 0.0894)							
Metrics	**Jaccard (↑)**	**PRAUC (↑)**	**F1 (↑)**	**DDI rate (↓)**	**Δ_{sp} Jaccard (↓)**	**Δ_{sp} PRAUC (↓)**	**Δ_{sp} F1 (↓)**	**Δ_{sp} DDI rate (↓)**
RETAIN (NeurIPS'16)	0.4417 ± 0.0007	0.6842 ± 0.0003	0.5960 ± 0.0006	0.0793 ± 0.0005	0.0202 ± 0.0005	0.0195 ± 0.0020	0.0149 ± 0.0017	0.0336 ± 0.0059
LEAP (KDD'17)	0.4302 ± 0.0005	0.5721 ± 0.0066	0.5830 ± 0.0004	0.0618 ± 0.0057	0.0435 ± 0.0082	0.0259 ± 0.0069	0.0356 ± 0.0076	0.0148 ± 0.0019
GAMENet (AAAI'19)	0.4854 ± 0.0010	0.7324 ± 0.0003	0.6350 ± 0.0010	0.0765 ± 0.0001	0.0205 ± 0.0019	0.0216 ± 0.0014	0.0157 ± 0.0018	0.0148 ± 0.0058
SafeDrug (IJCAI'21)	0.4628 ± 0.0008	0.7001 ± 0.0020	0.6149 ± 0.0006	**0.0587 ± 0.0006**	0.0322 ± 0.0069	0.0272 ± 0.0069	0.0245 ± 0.0056	0.0069 ± 0.0008
COGNet (WWW'22)	0.4954 ± 0.0032	0.7141 ± 0.0012	0.6442 ± 0.0031	0.0808 ± 0.0025	0.0274 ± 0.0012	0.0213 ± 0.0009	0.0198 ± 0.0011	0.0302 ± 0.0002
MoleRec (WWW'23)	0.4744 ± 0.0020	0.7136 ± 0.0012	0.6261 ± 0.0016	0.0713 ± 0.0008	0.0266 ± 0.0059	0.0239 ± 0.0065	0.0178 ± 0.0038	0.0093 ± 0.0011
RAREMed (SIGIR'24)	0.4768 ± 0.0003	0.7425 ± 0.0001	0.6261 ± 0.0002	0.0705 ± 0.0005	0.0327 ± 0.0002	0.0222 ± 0.0118	0.0242 ± 0.0003	0.0164 ± 0.0003
VITA (AAAI'24)	0.4919 ± 0.0021	0.6966 ± 0.0028	0.6368 ± 0.0017	0.0763 ± 0.0030	0.0162 ± 0.0025	0.0175 ± 0.0035	0.0143 ± 0.0023	0.0051 ± 0.0016
SubRec (NeurIPS'25)	0.4798 ± 0.0004	0.7233 ± 0.0002	0.6311 ± 0.0043	0.0692 ± 0.0015	0.0314 ± 0.0034	0.0243 ± 0.0021	0.0238 ± 0.0034	0.0155 ± 0.0035
HI-DR (AAAI'25)	0.6414 ± 0.0007	0.7916 ± 0.0019	0.7346 ± 0.0005	0.0835 ± 0.0001	0.0203 ± 0.0021	0.0133 ± 0.0023	0.0150 ± 0.0020	0.0038 ± 0.0011
RaceMED	**0.7157 ± 0.0015**	**0.8326 ± 0.0031**	**0.7890 ± 0.0020**	0.0899 ± 0.0017	**0.0155 ± 0.0031**	**0.0092 ± 0.0015**	**0.0100 ± 0.0023**	**0.0029 ± 0.0003**
Improvements (%)	**11.58**	**5.18**	**7.41**	-	**23.65**	**30.83**	**30.07**	-

Table 6: The effects of RaceMED's two core ideas: dual-branch encoder and race-aware attention mechanism on MIMIC-III

Dataset	MIMIC-III (Ground-Truth DDI Rate: 0.0815)							
Metrics	**Jaccard (↑)**	**PRAUC (↑)**	**F1 (↑)**	**DDI rate (↓)**	**Δ_{sp} Jaccard (↓)**	**Δ_{sp} PRAUC (↓)**	**Δ_{sp} F1 (↓)**	**Δ_{sp} DDI rate (↓)**
RaceMED	**0.7595 ± 0.0009**	**0.8803 ± 0.0007**	**0.8334 ± 0.0009**	**0.0825 ± 0.0009**	**0.0188 ± 0.0027**	**0.0171 ± 0.0036**	**0.0131 ± 0.0022**	0.0074 ± 0.0008
V1	0.7375 ± 0.0012	0.8661 ± 0.0010	0.8190 ± 0.0015	0.0843 ± 0.0005	0.0243 ± 0.0022	0.0204 ± 0.0031	0.0157 ± 0.0029	0.0057 ± 0.0009
V2	0.7351 ± 0.0009	0.8658 ± 0.0014	0.8169 ± 0.0012	0.0829 ± 0.0006	0.0234 ± 0.0020	0.0185 ± 0.0035	0.0147 ± 0.0032	**0.0050 ± 0.0015**
V3	0.7237 ± 0.0010	0.8652 ± 0.0011	0.8091 ± 0.0008	0.0839 ± 0.0009	0.0446 ± 0.0033	0.0269 ± 0.0029	0.0349 ± 0.0076	0.0095 ± 0.0011
V4	0.7385 ± 0.0007	0.8719 ± 0.0011	0.8195 ± 0.0011	0.0846 ± 0.0006	0.0275 ± 0.0027	0.0227 ± 0.0030	0.0216 ± 0.0024	0.0055 ± 0.0008

Table 7: The effects of RaceMED's two core ideas: dual-branch encoder and race-aware attention mechanism on MIMIC-IV

Dataset	MIMIC-IV (Ground-Truth DDI Rate: 0.0894)							
Metrics	**Jaccard (↑)**	**PRAUC (↑)**	**F1 (↑)**	**DDI rate (↓)**	**Δ_{sp} Jaccard (↓)**	**Δ_{sp} PRAUC (↓)**	**Δ_{sp} F1 (↓)**	**Δ_{sp} DDI rate (↓)**
RaceMED	**0.7157 ± 0.0015**	**0.8326 ± 0.0031**	**0.7890 ± 0.0020**	0.0899 ± 0.0017	**0.0155 ± 0.0031**	**0.0092 ± 0.0015**	**0.0100 ± 0.0023**	0.0074 ± 0.0003
V1	0.7050 ± 0.0019	0.8298 ± 0.0022	0.7821 ± 0.0018	0.0916 ± 0.0015	0.0203 ± 0.0023	0.0132 ± 0.0011	0.0136 ± 0.0031	0.0018 ± 0.0006
V2	0.7052 ± 0.0009	0.8279 ± 0.0017	0.7813 ± 0.0021	0.0922 ± 0.00015	0.0195 ± 0.0026	0.0139 ± 0.0018	0.0122 ± 0.0015	0.0021 ± 0.0015
V3	0.7042 ± 0.0021	0.8262 ± 0.0027	0.7779 ± 0.0019	0.0846 ± 0.0008	0.0446 ± 0.0025	0.0269 ± 0.0007	0.0349 ± 0.0019	0.0095 ± 0.0004
V4	0.7065 ± 0.0021	0.8215 ± 0.0030	0.7783 ± 0.0025	**0.0826 ± 0.0011**	0.0266 ± 0.0021	0.0192 ± 0.0022	0.0199 ± 0.0026	**0.0012 ± 0.0004**



(a) HI-DR (b) Race-General (c) Race-Specific (d) Final

Figure 6: Visualization of patient representations learned by (a) HI-DR, (b) RaceMED's race-general branch, and (c) RaceMED's race-specific branch, as well as (d) RaceMED's final representations. Colors and markers denote different racial groups.